



ACADEMIC REGULATIONS & SYLLABUS

Faculty of Science

**B Sc Programme
2022**



CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Education Campus – Changa, (ECC), hitherto a conglomerate of institutes of professional education in Engineering, Pharmacy, Computer Applications, Management, Applied Sciences, Physiotherapy, Nursing and Paramedical Sciences, is one of the choicest destinations by students. It has been transformed into **Charotar University of Science and Technology (CHARUSAT)** through an Act by Government of Gujarat. CHARUSAT is permitted to grant degrees under Section-22 of UGC-Govt. of India.

The journey of CHARUSAT started in the year 2000, with only 240 Students, 4 Programmes, one Institute and an investment of about Rs.3 Crores (INR 30 million). At present there are seven different institutes falling under ambit of six different faculties. The programmes offered by these faculties range from undergraduate (UG) to Ph.D. degrees. These faculties, in all offer 60+ different programmes. A quick glimpse is as under:

Faculty	Institute	Programmes Offered
Faculty of Technology & Engineering	Charotar Institute of Technology	B. Tech M. Tech MTM Ph. D
Faculty of Pharmacy	Ramanbhai Patel College of Pharmacy	B. Pharm M. Pharm MPM PGDCT/PGDPT Ph. D
Faculty of Management Studies	Indukaka Ipcowala Institute of Management	M.B.A PGDM Dual Degree BBA+MBA Ph.D
Faculty of Computer Applications	Smt. Chandaben Mohanbhai Patel Institute of Computer Applications	M.C.A/MCAL M.Sc (IT) Dual Degree BCA+MCA Ph. D
Faculty of Science	P D Patel Institute of Applied Sciences	M.Sc Dual Degree B.Sc+M.Sc Ph.D
Faculty of Medical Sciences	Ashok and Rita Institute of Physiotherapy	B.PT M.PT

	ManikakaTopawala Institute of Nursing	Ph.D B.Sc (Nursing) M.Sc
	Bapubhai Desaibhai Patel Institute of Paramedical Sciences	PGDHA BOP PGDMLT GNM Ph.D

The development and growth of the institutes have already led to an investment of over Rs.125 Crores (INR 1250 Million). The future outlay is planned with an estimate of Rs.250 Crores (INR 2500 Million). The University is characterized by state-of-the-art infrastructural facilities, innovative teaching methods and highly learned faculty members. The University Campus sprawls over 105 acres of land and is Wi-Fi enabled. It is also recognized as the Greenest Campus of Gujarat.

CHARUSAT is privileged to have 400+ core faculty members, educated and trained in IITs, IIMs and leading Indian Universities, and with long exposure to industry. It is also proud of its past students who are employed in prestigious national and multinational corporations.

From one college to the level of a forward-looking University, **CHARUSAT** has the vision of entering the club of premier Universities initially in the country and then globally. **High Moral Values like Honesty, Integrity and Transparency** which has been the foundation of ECC continues to anchor the functioning of **CHARUSAT**. Banking on the world class infrastructure and highly qualified and competent faculty, the University is expected to be catapulted into top 20 Universities in the coming five years. In order to align with the global requirements, the University has collaborated with internationally reputed organizations like Pennsylvania State University – USA, University at Alabama at Birmingham – USA, Northwick Park Institute –UK, Slovak Academy of Sciences, Slovakia, University of Bielsko-Biala, Poland, ISRO, BARC, Zydus Research Centre, etc.

CHARUSAT has designed curricula for all its programmes in line with the current international practices, National Education Policy 2020 and emerging requirements. Industrial Visits, Study Tours, Expert Lectures and Interactive IT enabled Teaching Practice form an integral part of the unique **CHARUSAT** pedagogy. The programmes are credit-based and have continuous evaluation as an important feature.

The pedagogy is student-centred, augurs well for self-learning and motivation for enquiry and research, and contains innumerable unique features like:

- The curriculum is contemporary and includes feature necessary to impart students the knowledge and skills necessary for undertaking career in academics, research, industry and entrepreneurship. Specialized courses also equip students with life-skills.
- Participatory and interactive discussion-based classes with extensive use of ICT tools.
- Sessions by visiting faculty members drawn from leading academic institutions, research organizations and industry.
- Regular weekly seminars and presentations.
- Distinguished lecture series, endowment chair talks, specialized interactive sessions with eminent academicians, researchers and entrepreneurs.
- Practical, field-based projects and assignments.

- Summer training in leading organizations under faculty supervision in relevant programmes.
- Industrial tours and field visits.
- Dissertation projects
- Fellowships for both undergraduate and post graduate students for exposure to research.
- Final Placement through campus interviews.
- Collaborations with research institutes, academic institutions, industries, industry associations and NOGs
- Effective Mentor-Mentee system
- Personalized counselling by expert psychologists

Exploration in the field of knowledge through research and development and comprehensive industrial linkages will be a hallmark of the University, which will mould the students for global assignments through technology-based knowledge and critical skills. The evaluation of the student is based on grading system. A student has to pursue his/her programme with diligence for scoring a good Cumulative Grade Point Average (CGPA) and for succeeding in the chosen profession and life.

CHARUSAT Welcomes You for A Bright Future



CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
Faculty of Science

ACADEMIC REGULATIONS

B Sc Programmes in Biological Sciences

Charotar University of Science and Technology (CHARUSAT)

CHARUSAT Campus, At Post: Changa – 388421, Taluka: Petlad, District: Anand

Phone: 02697-247500, Fax: 02697-247100, Email: info@charusat.ac.in

www.charusat.ac.in

2022

CHARUSAT

FACULTY OF SCIENCE

ACADEMIC RULES AND REGULATIONS [Bachelor's Degree (BSc) Programme in Biological Sciences]

The following are the academic rules and regulations for the **under graduate programmes in biological sciences** wherein the duration of the course, eligibility criteria and mode of admission, credit requirement and system of examination and other related aspects are laid down.

1. Nomenclature of the degree

The programme shall be called **Bachelor of Science (B Sc)** with specialization in Biotechnology, Microbiology and Biochemistry.

2. System of Education

- The Semester system and choice based credit system of education shall be followed.
- Duration of each semester will be of at least **90** working days.
- Every student requires to do a specified course work in the chosen program of specialization and also to complete project/dissertation/industrial training/specialized training if any. Courses have been structured adhering to the concepts of Outcome Based Education (OBE).
- **Medium of instruction will be English.**

3. Duration of Programme

- Undergraduate programme (B Sc) 6 semesters (3 academic years) and B Sc (Honours) 8 Semesters (4 academic years).
- Maximum time allowed for completion of the B Sc programme shall not be more than 12 semesters for B Sc and 16 semesters for B Sc (Honours), respectively.

4. Eligibility for admission

For the admission to BSc programs in Biological Sciences, a candidate must have successfully passed the higher secondary examination with science subjects from a state or central or international school education boards etc.

5. Mode of admission

Admission to B Sc programme will be based on merit.

6. Course structure and credit scheme

A student admitted to the program should study the prescribed courses and earn credits specified in the course structure. Courses in a programme will be of five major kinds:

(a) Core, (b) Discipline Specific Electives, (c) Ability Enhancement Compulsory Courses, (d) Skill Development Courses and (e) University Elective courses. These courses can be further segregated as follows:

- (a) **Core courses (Theory and Laboratory):** These are the courses which should compulsorily be studied by a candidate and are called “Core compulsory”.
- (b) **Discipline Specific elective:** These are the courses where choice or an option for the candidate is available. The student can choose the courses from the sister/allied disciplines which supports the main discipline.
- (c) **Ability Enhancement Compulsory Course (AECC):** These are the courses based upon the content that leads to knowledge enhancement
- (d) **Skill Development Course (SEC):** These are value-based and/or skill-based and are aimed at providing hands-on-training, competencies, skills, etc
- (e) **Foundation courses:** These are the courses based upon the content that leads to knowledge enhancement and/or skill enhancement. These are mandatory for all disciplines but choices are available to the candidate from the defined courses.
- (f) **University Electives:** These are diverse courses of interdisciplinary nature offered by different faculties/centres of the University
- (g) **Additional/Value-added credits:** NSS/NCC/Sports/Cultural Activities/ Yoga & Wellness/ Seminar/Industrial tour/Field visit/Summer internship
- (h) **Project work:** Special course involving application of knowledge in solving / analyzing /exploring a real-life situation / difficult problem.
- (i) **Vocational Training:** Vocational training imparted in association with Board of Apprenticeship Training (BOAT).

The courses offered during the first two semesters are basic and introductory courses in Biology, Chemistry, Computer applications, and Science for Community Development. These are mandatory for all the students and are meant to give a flavour to the various approaches and analyses used in these disciplines and to prepare the students for advanced courses in later years of study. In addition, there will be Interdisciplinary as well as Trans-disciplinary Courses. These courses aim at (i) imparting communication skills, computational skills and information literacy, mathematical and statistical skills, analytical skills (ii) developing abilities like scientific inquiry, critical thinking and problem solving, self-expression through liberal arts and languages (iii) preparing students for life-long learning by inculcating holistic competencies like professional ethics, values of culture and wisdom, social responsibilities, leadership and global competency.

From the third semester onwards, students have the freedom to choose advanced courses in their choice of specialization i.e. biotechnology, microbiology and biochemistry based on their interest and inclination, within the institutional guidelines for branch selection. In addition, these students are also exposed to industrial training and field investigations. By availing this, a student can work in an experimental lab or visit industries or environmental sites and biodiversity spots. This is meant to help the student to gain required personality traits for working in an industrial environment while developing environmental consciousness.

Credit scheme

- The B Sc programme shall have **21-25 credits/semester** and the total minimum **credits to earn a B Sc degree is 138 while for B Sc (Hons) 147 credits are needed.**
- A candidate has to register for minimum prescribed credits in a semester.

7. Attendance

- All activities prescribed under these regulations and listed by the course faculty members in their respective course outlines are compulsory for all students pursuing the courses.
- Minimum attendance required is 80%.

8. Course Evaluation

The performance of every student in each course will be evaluated as follows:

- Continuous and Comprehensive assessment is an integral part of CBCS of CHARUSAT. A continuous assessment system in semester system (also known as internal assessment/comprehensive assessment) is spread through the duration of course and is done by the teacher(s) teaching the course.
- Internal evaluation by the course faculty member(s) based on continuous assessment for 30% of the marks for the course; and final examination 70% of the marks for the course.
- The assessment is done through various means including: Written tests, MCQ based quizzes, Presentations, Mini-projects, Projects, dissertation, Field visits, industry visits, Seminars, assignments, group discussions/activities, Viva-voce, Online tests etc. as prescribed by the concerned teacher and decided by the Faculty of Science **(Refer to annexure-I)**
- In “Continuous evaluation” of practical students shall be evaluated in a continuous manner for their involvement in the practical, aptitude for learning, completion of practical related assignments, regularity in the laboratory, skills acquired, critical analysis of experimental data, effective representation of data through written and presentation mode, and record keeping.
- The end examination for 70% of the evaluation for the course will be through written paper or practical test or oral test or presentation by the student or online tests or a combination of any two or more of the above.
- In order to earn the credit in a course a student has to obtain grade other than FF.
- Remedial exams may be conducted after fifteen days of declaration of result of University Winter/Summer Examination. Also, one more June-Supplementary exam shall be conducted in the last week of June (end of Academic year) for the students who are left from odd / even remedial University Exam.

9. Grading

Grading Scheme: Relative grading based on ten-point scale will be adopted.

Letter Grade	AA	AB	BB	BC	CC	CD	DD	FF
Grade Point	10	9	8	7	6	5	4	0

The class awarded to a student in the programme is decided by the final CGPA

- The student's performance in any semester will be assessed by the Semester Grade Point Average (SGPA). Similarly, his performance at the end of two or more consecutive semesters will be denoted by the Cumulative Grade Point Average (CGPA). The SGPA and CGPA are calculated as follows:

- SGPA = $\sum C_i G_i / \sum C_i$ where C_i is the number of credits of course i
 G_i is the Grade Point for the course i and $i = 1$ to n ,
 n = number of courses in the semester
- CGPA = $\sum C_i G_i / \sum C_i$ where C_i is the number of credits of course i
 G_i is the Grade Point for the course i and $i = 1$ to n ,
 n = number of courses of all semesters up to which CGPA is computed.
- In addition, the student has to complete the required formalities as per the regulatory bodies.

In case, a student, of undergraduate programme or diploma, gets less than 40% marks in end-semester examination and less than 45% marks overall (combining continuous evaluation and end-semester examination) in a particular course, he / she will not be graded in that course till he / she reappears in said course and obtains specified minimum marks. **A student will not be allowed to move to next year if she/he has not cleared all the courses of previous year.**

10. Grading Procedure

The committee will be constituted for finalising the Grading.

- Provost Nominee
- Dean
- Expert Member(s)

11. Award of Degree

Every student will be eligible for the award of the degree if she/he **should have earned minimum required credits** as prescribed in a program.

12. Provision for Appeal

There shall be a provision for Appeal for a candidate who may be dissatisfied with the Grade he/she has been awarded. He/she can approach the University with the written submission.

13. Award of Class

The class awarded to a student in the programme is decided by the final CGPA as per the following scheme:

Award of Class	CGPA Range
First Class with Distinction	$\text{CGPA} \geq 7.50$
First class	$7.50 > \text{CGPA} \geq 6.50$
Second Class	$6.50 > \text{CGPA} \geq 5.50$
Pass Class	$5.50 > \text{CGPA} \geq 4.00$

14. Transcript

The transcript will be issued to the student only after successful completion of the course by earning required number of credits for B.Sc. degree. Transcript will contain a consolidated record of all the courses taken, credits earned, grades obtained, SGPA, CGPA, class obtained, etc.

Annexure-I
Policy for distribution of Internal marks

Particulars	Marks	Marks distribution for 2 credit course	Marks distribution for 3 or 4 credit course
All the internal test/other continuous assessment tool	15 / 30 Marks Sum of average marks obtained by the student from internal test + marks of other assessment components	10 (Should be converted in to 10 at the end of semester)	20 (Should be converted in to 20 at the end of semester)
Assignment Or Seminar	3 /Course	3	6
Attendance	2/Course	2	4
Total		15	30

Charotar University of Science and Technology
P. D. Patel Institute of Applied Sciences
BACHELOR OF SCIENCE

VISION

To become a pre-eminent national institute imparting science education integrated with research.

MISSION

To engage in education, research, and spread of science for the benefit of society.

PROGRAM EDUCATIONAL OBJECTIVES

The Graduate would:

PEO 1: Recognize opportunities to enhance their knowledge and skills as a lifelong process.

PEO 2: Possess scientific aptitude and competency in execution of assigned tasks,

PEO 3: Have ability to understand and contribute towards solving global problems.

PEO 4: Understand the importance of team work and possess leadership and entrepreneurship traits.

PROGRAM OUTCOMES (POs)

The Programme outcome are as under:

PO1. Knowledge in sciences: Possess knowledge and comprehend the various core and allied courses offered under the opted stream of sciences.

PO2. Planning Abilities: Demonstrate effective planning abilities, including time management, delegation skills, organizational skills, teamwork, and interpersonal skills.

PO3. Problem analysis: Utilize the principles of scientific enquiry and think analytically, critically with clarity of thought, while solving problems.

PO4. Modern tool usage: Apply appropriate methodologies, methods and procedures, resources to conceptualize the matters.

PO5. Leadership skills: Realize the importance of teamwork and take initiatives to plan and implement a workflow to fulfil tasks with a compassionate attitude.

PO6. Professional Identity: Establish as a professional in the chosen area of the profession with social responsibility.

PO7. Ethics in Science: To understand the value of being ethical and implement its principles while executing tasks at workplace and in life. To respect and adhere to the rules/regulations/treaties/directives/notifications/laws related to scientific research and business.

PO8. Communication: Communicate effectively and responsibly, comprehend and write reports, make presentations, keep records and documents, and give and receive clear instructions.

PO9. Modern Science and Society: Apply reasoning informed by the contextual knowledge to assess societal, health, safety and legal issues and help society whenever situations arise.

PO10. Environment and sustainability: Understand the causes of environmental pollution and how it can be reduced and remedied using green technology in industries and environment.

PO11. Life-long learning: Vigilant about changes in the worldwide professional scenario arising out of technological, political, and economic factors. Recognize the need for and undertake steps necessary to fill the gaps recognized to keep oneself professionally competent. Self- assess and use feedback effectively from others to identify learning needs and to satisfy these needs on an ongoing basis.

Teaching scheme for B.Sc. Biotechnology/Microbiology/Biochemistry

Key to Course Codes

BT: B Sc (Biotechnology) specific courses

MI: B Sc (Microbiology) specific courses

BC: B Sc (Biochemistry) specific courses

BS: Common courses for all three branches

BE: Common elective courses for the all the three branches

B. Sc. Biotechnology/Microbiology/Biochemistry (Semester I)

Course Type	Semester - I							Theory Evaluation			Practical Evaluation			Grand Total (I+II)
	Course Code	Course Title	L	P	T	Contact Hours	Total Credits	Institute	University	Total-I	Institute	University	Total-II	
Core Course (Theory)	BS118	Chemistry	3	-	-	3	3	30	70	100	-	-	-	100
	BS119	Animal Biology	3	-	-	3	3	30	70	100	-	-	-	100
	BS120	Plant Biology	3	-	-	3	3	30	70	100	-	-	-	100
	BS121	Ecology and Environment	2	-	-	2	2	15	35	50	-	-	-	50
Core Course (Laboratory)	BS122	Laboratory in Biological Sciences-I	-	12	-	12	6	-	-	-	50	100	150	150
Discipline specific Elective (DSE)														
Ability Enhancement Compulsory Course (AECC)	HS101.02E	Communicative English	-	2	-	2	2	-	-	-	30	70	100	100
	FS102 E	Foundation Course on Chemistry and Biology	-	2	-	2	2				50	50	100	100
Skill Development Course (SEC)		-	-	-	-	-	-	-	-	-	-	-	-	-
University Elective		-	-	-	-	-	-	-	-	-	-	-	-	-
Additional/Value-added credits*		-	-	-	-	-	-	-	-	-	-	-	-	-
Total			11	16		27	21	-	-	-	-	-	-	700

B.Sc. Biotechnology/Microbiology/Biochemistry (Semester II)

	Semester - II							Theory Evaluation			Practical Evaluation			Grand Total (I+II)
Course Type	Course Code	Course Title	L	P	T	Contact Hours	Total Credits	Institute	University	Total-I	Institute	University	Total-II	
Core Course (Theory)	BS218	Principles of Biochemistry	3	-	-	3	3	30	70	100	-	-	-	100
	BS219	Fundamentals of Microbiology	3	-	-	3	3	30	70	100	-	-	-	100
	BS220	Tools and Techniques in Biology	3	-	-	3	3	30	70	100	-	-	-	100
	BS221	Foundations of Biotechnology	2	-	-	2	2	15	35	50	-	-	-	50
Core Course (Laboratory)	BS222	Laboratory in Biological Sciences- II	-	12	-	12	6	-	-	-	50	100	150	150
Discipline specific Elective (DSE)														
Ability Enhancement Compulsory Course (AECC)	HS201.02E/HS202.02E/HS205.02E/HS209.02E/HS210.02E	Liberal arts	-	2	-	2	2	-	-	-	30	70	100	100
	FS101E	Foundation Course on Mathematics and Physics	-	2	-	2	2				50	50	100	100
Skill Development Course (SEC)		-	-	-	-	-	-	-	-	-	-	-	-	-
University Elective		-	-	-	-	-	-	-	-	-	-	-	-	-
Additional/Value-added credits*		-	-	-	-	-	-	-	-	-	-	-	-	-
Total			11	16		27	21	-	-	-	-	-	-	700

B.Sc. Semester I Detailed Syllabus

BS118 CHEMISTRY

Semester I

Credits and Hours:

Teaching scheme	Theory	Practical	Total	Credit
Hours/week	3	-	3	3
Marks	100	-	100	

A. Outline of the course

Sr. No.	Title	Minimum number of hours
1.	General Chemistry	11
2.	Basics of Inorganic Chemistry	12
3.	Conceptual Organic Chemistry	10
4.	Fundamentals of Physical Chemistry	12

B. Instructional Methods and Pedagogy

Conventional blackboard as well as power-point presentation will be the medium for discussion of topics. More emphasis will be given to the interactive classroom session. Structural models will be provided for better understanding of the structural chemistry and geometry of the molecules. Several numerical problem-solving sessions will be conducted for better understanding of the subject. Along with the text book knowledge, students will be enriched by providing the course materials. Internal exams and surprise quizzes will be conducted regularly as a part of continuous evaluation and suggestions will be given to student in order to improve their performance. Practical sessions are designed in such a way so that every students can perform their experiments in a well equipped chemistry laboratory. The practical experiments will enhance the hands-on ability performance.

C. Detailed Syllabus:

Sr. No.	Title	Minimum number of hours
1.	General Chemistry: Atomic structure and The Periodic Table: Bohr's model and its limitations, de Broglie's relationship, Heisenberg uncertainty principle, concept of orbitals, quantum numbers, shapes of s, p and d orbitals, rules for filling electrons in orbitals - Aufbau principle, Pauli's exclusion principle and Hund's rule, electronic configuration of atoms, Periodic trends in properties of elements - atomic radii, ionic radii, inert gas radii, Ionization energy, electronegativity, valency. ii) Chemical Bonding and Molecular Structure: Covalent bonding, ionic bonding, Valence Bond Theory (VBT), VSEPR	11

	theory, Hybridization, Molecular Orbital Theory (simple homo and hetero nuclear molecules). iii) Radioactivity and Nuclear Chemistry: Radioactive decay (alpha, beta and gamma radiation), Radioisotopes and its application, radioactive half -lives, nuclear fission and fusion.	
2.	Basics of Inorganic Chemistry: i) First Row Transition Elements (3d) and Coordination Chemistry: General group trends with special reference to electronic configuration, variable valency, Colour, magnetic properties, Metals in biological system Crystal Field Theory (CFT), Crystal field stabilization energy (CFSE), Ligand Field Theory (Strong and Weak field), Octahedral and Tetrahedral geometry, Jahn-Teller distortion. ii) Acids and Bases: Arrhenius theory of Acids and Bases, The Lowry – Bronsted Concept, Strength of Acids and Bases, The Lewis concept, Ionization of weak acids and bases in aqueous solution, ionization constants, ionic product of water, pH-scale. iii) Redox Chemistry and Electrochemical Cell: Concept of oxidation and reduction, redox reactions, oxidation number, balancing redox reactions - in terms of loss and gain of electrons and change in oxidation number. Electro chemical cells, half-cell, electrodes potential standard electrode potential, Nernst equation, redox potential, reference electrode, standard cell, battery, corrosion.	12
3.	Conceptual Organic Chemistry: i) Amino Acids and Peptides: Classification of Amino Acids, Zwitter-ion structure, Determination of primary structure of peptides, determination of N-terminal amino acid and C-terminal amino acid, Synthesis of simple peptides. ii) Stereochemistry: Projection Structures of Stereoisomers (Fischer, Sawhorse, Newman, Flying-Wedge projection), Enantiomers, Diastereomers, Different types of Isomerism, Structural Isomers, Geometrical, Stereoisomerism, Configurational Isomers, Conformational Isomers, Stereochemical nomenclature of Stereoisomers containing chiral centers(R/S and E/Z or cis-trans, D,L system of designation. iii) Substitution and Elimination Reactions: Nucleophilic substitution reactions (S_N1 and S_N2 reactions), Electrophilic Substitution Reactions, Eliminations reactions (E_1 and E_2 reactions), elimination vs substitution.	10

4.	Fundamentals of Physical Chemistry: i) Thermodynamics: Concepts of System and types of systems, surroundings, work, heat, energy, extensive and intensive properties, state functions. <i>First law of thermodynamics:</i> Cyclic process, Reversible & irreversible process, internal energy, enthalpy, work done an isothermal & adiabatic process, heat capacities. <i>Second law of thermodynamics:</i> Carnot cycle, Elementary treatment of entropy, free entry, work function & criterion of equilibrium. Gibbs Helmholtz equation, <i>Third law of thermodynamics</i> (brief introduction). ii) Homogeneous chemical equilibrium: Law of mass action and equilibrium constant K_p, K_c, K_x and their relationship. Le-chatelier principal- effect of temperature, pressure and addition of products of relation and inert gases. Vant's hoff equation (derivation not required) and its application. iii) Introduction to spectroscopy: Electromagnetic radiation, fundamental definitions, electromagnetic spectrum and different spectroscopy, introduction to concepts of absorption and emission spectroscopy, Vibrational and Electronic spectroscopy, Beer-Lambert law.	12
----	--	----

Total Hours: 45

D. Reference Books:

- James E. Huheey, "*Inorganic Chemistry: Principles of structure and reactivity*", Prentice Hall, IV Edition.
- D. S. Shriver and P.A. Atkins, "*Inorganic Chemistry*", Oxford University Press, IV Edition.
- J. D. Lee, "*A New Concise Inorganic Chemistry*", ELBS IV Edition
- I. L. Finar: *Organic Chemistry* (Vol. I & II), E. L. B. S.
- T. W. Graham Solomon's *Organic Chemistry*, John Wiley and Sons.
- Peter Sykes: *A Guide Book to Mechanism in Organic Chemistry*, Orient Longman.
- D. Nasipuri, *Stereochemistry of Organic Compounds*, New Age International Publishers.
- P.S. Kalsi, *Stereochemistry, Conformation and Mechanism*, John Wiley and Sons.
- R. P. Sarkar, *General and Inorganic Chemistry: Volume I*
- I. Levine, *Physical Chemistry*.
- P. C. Rakshit, *Physical Chemistry*.

E. Course Outcomes: The students will be able to comprehend:

- CO1: The basic understanding of chemistry to the undergraduate students of biological sciences.
- CO2: Correlation between chemistry and biology and how chemistry is involved in biological system
- CO3: Apply the basic principles chemistry at the chemistry laboratory.

BS119 ANIMAL BIOLOGY
Semester I

Credits and Hours:

Teaching scheme	Theory	Practical	Total	Credit
Hours/week	3	-	3	3
Marks	100	-	100	

A. Outline of the Course

Sr. No.	Title	Minimum number of hours
1.	Animal evolution and Diversity	12
2.	Animal Behaviour	9
3.	Animal Physiology	12
4.	Reproduction and Development Biology	12

B. Methodology & Pedagogy

The topics will be discussed in interactive class room sessions using classical black-board teaching to power-point presentations. Practical sessions will be conducted in a highly equipped laboratory. Experiments will be carried out by students individually. Special interactive problem-solving sessions will be also conducted by respective faculty members. Course materials will be provided to the students from various primary and secondary sources of information. Internal exams will be conducted regularly as a part of continuous evaluation and student will be counselled to improve their performance.

C. Detailed syllabus

Sr. No.	Title	Minimum number of hours
1.	Animal evolution and Diversity History of life on earth, Species and Speciation, Darwin's Natural Selection Theory, evolution of eukaryotic cell from prokaryotic cell, endosymbiotic theory; Principles of Systematics and animal classification in Chordates and Non-chordates with characters and examples	12
2.	Animal Behaviour Introduction of Ethology, Branches, History, Significance, Patterns of Behaviour, Study of Behaviour, Genetics of Behaviour, Reproductive Behaviour	9

3.	Animal Physiology Tissues, Organs and Systems, Feeding and Digestion, Absorption, Metabolism, Respiration, The Blood and Circulatory System, Excretion, Muscle physiology, Nervous System, Hormones and Endocrine glands	12
4.	Reproduction and Development Biology Reproduction in selected vertebrates and invertebrates; Basic concepts of embryonic development; gametogenesis, Pre and post embryonic developments in human beings	12

Total Hours: 45

D. Recommended study materials

- Gilbert S.F. Developmental Biology, 10th Edition, Sinauer Associates, Inc., Publishers Sunderland, Massachusetts, USA.
- Textbook of Invertebrates by RL Kotpal, Rastogi Publication, Meerut.
- Textbook of Vertebrates, RL Kotpal, Rastogi Publications, Meerut.
- Chordate Zoology, PS Dhami and JK Dhami, S Chand & Co., Delhi.
- Invertebrate Zoology, Jordan and Verma, S Chand & Company, Delhi
- Instant notes in Animal Biology by Richard D Jurd. Vivi Books Pvt Ltd, New Delhi.
- Evolution and Behaviour by ReenaMathur, SP Singh and BS Tomar 1st Ed., 2013 Rastogi Publication, Meerut.
- Evolutionary Biology by BS Tomar and SP Singh. 9th Ed. 2013-14, Rastogi Publications, Meerut.
- A Textbook of Animal Physiology by AK Berry, Emkay Publications, Delhi.
- Animal Physiology by VK Agarwal and PS Verma. Reprint Edition: (2012), S Chand &Co., Delhi.
- Animal Physiology by KA Goel and KV Sastry, 6th Ed 2012-13, Rastogi Publications, Meerut.
- Animal Physiology and Biochemistry by KV Sastry, 2nd Ed. 2012-13, Rastogi Publications, Meerut.
- Instant notes in Animal Biology by Richard D Jurd. Vivi Books Pvt Ltd, New Delhi.
- A Manual of Practical Zoology: Invertebrates, By: V.K. Aggarwal, Dr. P. S. Verma. S Chand and Co.
- Practical Zoology Invertebrates by Dr. S. S. Lal, 2013, Rastogi Publications.
- A Manual of Practical Zoology Chordates byDr. P. S. Verma, S Chand and Co.
- Practical Zoology vertebrates by SS Lal, Rastogi Publications.

E. Course Outcome: Upon successful completion of the course, student will learn about:

CO1: Systematics, evolution and animal classification

CO2: Various aspects of animal behavior.

CO3: Physiological processes occurring in animal systems

CO4: Basic developmental processes in animals and Process of reproduction in selected vertebrates and invertebrates

BS120 PLANT BIOLOGY**Semester I****Credits and Hours:**

Teaching scheme	Theory	Practical	Total	Credit
Hours/week	3	-	3	3
Marks	100	-	100	

A. Outline of the Course

Sr. No.	Title	Minimum number of hours
1.	Introduction to Systematics and morphogenesis	13
2.	Histology and Anatomy	10
3.	Physiology	12
4.	Cryptogams, Gymnosperms and Angiosperms	10

B. Methodology & Pedagogy

The topics will be discussed in interactive class room sessions using classical black-board teaching to power-point presentations. Practical sessions will be conducted in a highly equipped laboratory. Experiments will be carried out by students individually. Special interactive problem-solving sessions will be also conducted by respective faculty members. Course materials will be provided to the students from various primary and secondary sources of information. Internal exams will be conducted regularly as a part of continuous evaluation and student will be counselled to improve their performance.

C. Detailed syllabus

Sr. No.	Title	Minimum number of hours
1.	<i>Introduction to Plant Sciences, Systematics and morphogenesis</i> Introduction to plant sciences and its scope, Sub branches of botany, Plant Sciences and its relevance in present time. Taxonomy, concept of naming plants with examples and systematics in plant sciences, Systems of classification, their basis, periods and Development with importance of each classification system Plant Morphology: roots, stem, leaf, inflorescence, flower, pollination, fertilization, seeds, fruits, dispersion of seeds and fruits	13
2.	<i>Histology and Anatomy</i> Cell, tissue, tissue system, anatomy of root, leaves, stem, secondary growth	10

3.	Physiology Chemical composition of Plants and Mineral Nutrients, Absorption and Movements of water, Photosynthesis - Light harvesting complexes; mechanisms of electron transport; C3, C4 and CAM pathways. Respiration and photorespiration – Citric acid cycle; plant mitochondrial electron transport and ATP synthesis; alternate oxidase; photorespiratory pathway, Carbon Assimilation Nitrogen Metabolism and special modes of nutrition, Translocation and storage of food Respiration and Photorespiration Growth (Primary and secondary, Its importance, measurement of plant growth), Transpiration; Growth, hormones, enzymes and vitamins Movement, Reproduction, Translocation and storage of food; Stress, aging, senescence and Death; Induced systemic resistance and systemic acquired resistance; Seed germination- Hormonal control of seed germination, Vernalization and hormonal control	12
4.	Cryptogams, Gymnosperms and Angiosperms Crptogams: Types and introduction to Algae, bacteria, fungi, lichens. Life cycle of Bryophyta and Pteridophyta Gymnosperms: Any one of Cycadales or coniferal or Gnetales. Lifecycle and applications Angiosperms: Development of male and female gametes, pollination, fertilization in both Dicotyledons and monocotyledons (life cycle and applications) Introduction to Economic Botany, plant-based products and technology	10

Total Hours: 45

D. Recommended study materials

- Nagesh Ashish, Prashant Kumar and HussianKushal Life Sciences UGC CSIR NET/SET. Arihant publication ISBN: 9789350750483
- Dutta AC (2013) Botany for degree students. 6th Edition Oxford University Press ISBN: 13:978-0-819-563748-9
- Kumar A (2001) Botany in Forestry and Environment, 1st Edition. Avichal Press

E. Course Outcome:

Upon successful completion of the course, student will possess basic understanding of:

- CO1: Important aspects of plant sciences
- CO2: Taxonomy and systematic of plants.
- CO3: Various cell and tissue type of plants
- CO4: Growth and physiological factors influencing plants

BS121 ECOLOGY AND ENVIRONMENT
Semester I

Credits and Hours:

Teaching scheme	Theory	Practical	Total	Credit
Hours/week	2	-	2	2
Marks	50	-	50	

A. Outline of the Course

Sr. No.	Title	Minimum number of hours
1.	Introduction to Ecology	5
2.	Components and factors of Ecology	8
3.	Environmental pollution, prevention and SDGs	10
4.	Functions of ecosystem and Important ecosystem services	7

B. Methodology & Pedagogy

The topics will be discussed in interactive class room sessions using classical black-board teaching to power-point presentations. Practical sessions will be conducted in a highly equipped laboratory. Experiments will be carried out by students individually. Special interactive problem-solving sessions will be also conducted by respective faculty members. Course materials will be provided to the students from various primary and secondary sources of information. Internal exams will be conducted regularly as a part of continuous evaluation and student will be counselled to improve their performance.

C. Detailed syllabus

Sr. No.	Title	Minimum number of hours
1.	Introduction to ecology Introduction and history of ecology, branches of ecology, ecology and other disciplines, ecological tools and techniques, significance of ecology-ecosystem degradation and resource utilization	5
2.	Components and factors of Ecology Environment: Hydrosphere, lithosphere, water, soil, Environmental Factors: Biotic- Interspecific interactions: positive interaction-mutualism, commensalism, photo cooperation; and negative- exploitation, amensalism, competition Abiotic- climatic, topographic and edaphic Climatic: light temperature, water, atmosphere (gas and wind), fire Topographic: latitudes & altitude, height of mountain chains, direction of mountains and valleys, steepness of slope	8

3.	Environmental pollution, prevention and SDGs Types of environment pollution (Air, water, soil, marine, thermal, noise, nuclear pollution), Causes, Prevention and Sustainable Development Goals	10
4. 4.	Functions of ecosystem and Important ecosystem services Structure: Producers, consumers and decomposers Function: biogeochemical cycles (water, Carbon, oxygen, nitrogen), Energy flow in ecosystem: food chain, Energy cycles Types of ecosystems: Aquatic, terrestrial, marine, dessert Important ecosystem services. Population biotic communities and ecological successions	7

Total Hours: 30

D. Recommended study materials

- Verma PS and VK Agarwal (2012) Cell biology, genetics, molecular biology, evolution and ecology. Multicolor edition, S. Chand & company ISBN:81-219-2442-1
- Bharuch Erach (2008) Textbook of Environmental Studies for undergraduate courses, University Grant Commission, University Press ISBN: 13:978-81-7371-5402

E. Course Outcome:

Upon successful completion of the course, student will possess basic understanding of:

- CO1: basic understanding of ecological components
- CO2: Various biogeochemical cycles in the ecosystem.
- CO3: Factors affecting ecosystem
- CO4: Various ecological tools and techniques